

Wireless Hotel Ordering System

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Traditional ordering systems consist of waiters handing over the menu to customers and taking orders from them. The waiter's performance is defined by the speed at which the order gets processed, and it might get delayed at times.

In the wireless ordering system, there is no waiter to take the order. Everything related to the waiter is done by the microcontroller (MCU) and a wireless transmitter. The customer can start ordering as soon as he or she is seated comfortably at the table. The delay caused by the waiter is eliminated in this system.

Circuit and working

The wireless hotel ordering system consists of a transmitter and a receiver section. The block diagram of the transmitter unit is shown in Fig. 1 and the receiver unit in Fig. 2.

The complete circuit diagram of the transmitter is shown in Fig. 3 and of the receiver is shown in Fig. 4.

Microcontroller. The heart of this circuit is an 8-bit AVR ATmega8 MCU

that controls, stores and coordinates the activities of the system. The software program stored in the MCU controls the functions of the system.

AVR combines the most code-efficient architecture for C language and assembly programming with the ability to tune system parameters throughout its life cycle. ATmega8 provides the following features: 8k bytes of in-system programmable flash with read-write capabilities, 1k byte of SRAM, 23 general-purpose I/O lines, 32 general-purpose working registers, three flexible timers/counters with compare modes, internal and external interrupts, a serial programmable universal synchronous/asynchronous receiver/transmitter (USART), an SPI serial port and five software-selectable power saving modes.

The flash program memory can be reprogrammed in-system through an SPI serial interface by a conventional non-volatile memory programmer or an on-chip boot program running on AVR core. With AVR's USART you just need to write the data to one of the registers of USART and you are free to do other things, while USART is transmitting the bytes.

Test Points

Test point	Details
TP0, TP4	0V (GND)
TP1, TP5	5V
TP2-TP3	Train of pulse when S4 button is pressed
TP6-TP7	Train of pulse when S4 button is pressed

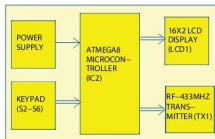


Fig. 1: Block diagram of the wireless hotel ordering system—transmitter side

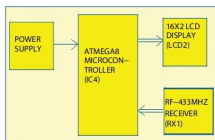


Fig. 2: Block diagram of the wireless hotel ordering system—receiver side

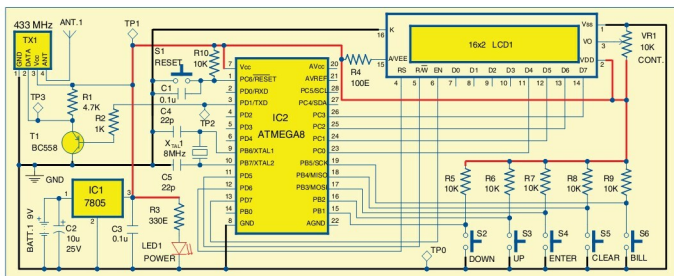


Fig. 3: Circuit diagram of the transmitter section